

Customer Lifetime Value (CLV) Forecasting and Segmentation

Project Report

Prepared from the CLV modeling notebook

Executive Summary

This project estimates customer lifetime value using transactional retail data and converts those estimates into practical customer segments for retention and growth planning. The workflow combines a probabilistic CLV model with a machine learning benchmark, then uses the results to identify high-value customers, understand spending patterns, and suggest commercial actions.

Problem Statement

The business needed a way to understand which customers are likely to generate the most value over time, which customers are at risk of low future spend, and where to focus retention or upsell efforts. A simple historical revenue view is not enough, because it does not capture future value or customer behaviour over time.

Data and Preparation

The notebook uses the Online Retail II transaction dataset. The data was combined from two files and then cleaned before modeling.

- Removed rows without Customer ID.
- Excluded cancelled invoices (invoice numbers starting with 'C').
- Removed zero or negative quantity rows.
- Converted Customer ID to integer type.
- Created TotalPrice as Quantity × Price.
- Parsed InvoiceDate into datetime format and used it for customer history calculations.

A snapshot date was created using the latest transaction date plus one day so that customer recency and tenure could be measured consistently.

Methodology

1) Probabilistic CLV modeling. The notebook used the lifetimes library with BG/NBD to model purchase frequency and Gamma-Gamma to estimate monetary value. These two models were combined to forecast six-month customer lifetime value.

2) Customer tiering. Predicted CLV values were split into quartiles to create Low-Tier, Mid-Tier, and Top-Tier customer segments.

3) Business validation. The notebook also built a holdout-based comparison and trained an XGBoost model to benchmark prediction accuracy against the probabilistic approach.

Key Results

Metric	Result
Top 10 customers forecast	£36k to £129k over the next 6 months
Total six-month revenue forecast	Approximately £4.5 million
Top-Tier average predicted CLV	£2,349
Mid-Tier average predicted CLV	£327
Low-Tier average predicted CLV	£82
Average spend across all customers	£580.08
Average spend among buyers	£1,335.40

CLV Tier Profile

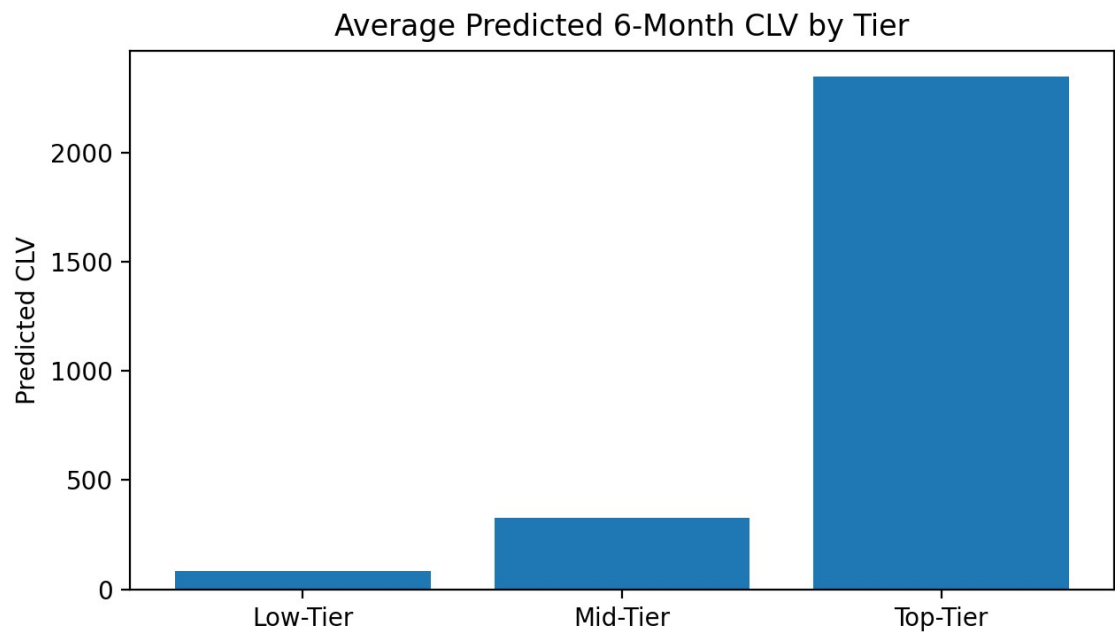


Figure 1. Average predicted 6-month CLV by customer tier.

Customer Profiling Insights

After merging CLV predictions back with transactional data, the notebook found that the United Kingdom is the dominant geography among top-value customers. The analysis also suggests that product mix differs by tier: low-value customers buy a broader mix, while top-tier customers show stronger preference for specific higher-value products.

- Focus marketing on mid-tier customers in the UK, where conversion to top-tier looks most realistic.
- Use product-level recommendations to push higher-margin or repeat-purchase categories.
- Treat top-tier customers as a priority retention cohort with loyalty or VIP-style offers.

Model Comparison

A second model, XGBoost, was trained on holdout spend to compare performance with the probabilistic model.

Model	RMSE	MAE
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XGBoost	2640.63	364.01
Lifetimes	3182.28	434.57

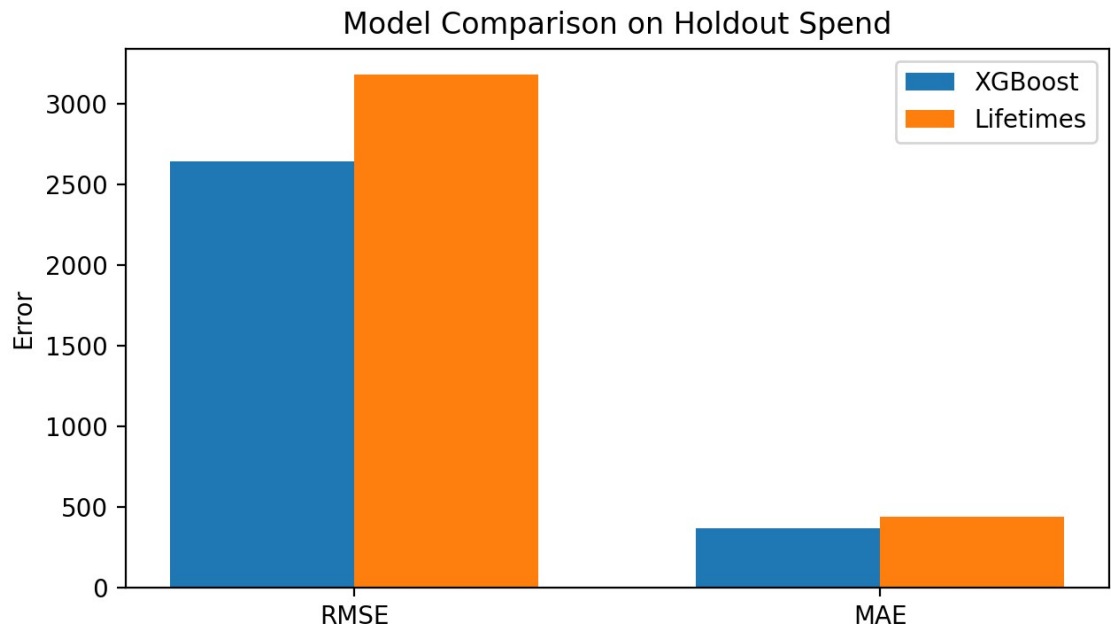


Figure 2. Holdout error comparison between XGBoost and the probabilistic model.

The benchmark shows that XGBoost is more accurate for tactical short-term forecasting, while the lifetimes model is more useful for strategic CLV estimation and segmentation, especially when customer history is limited.

Business Recommendations

- Prioritize retention campaigns for Top-Tier customers to protect the highest-value revenue base.
- Use Mid-Tier customers as the main growth pool, with targeted offers, category recommendations, and personalized messaging.
- Build geographic campaigns around the UK, where the strongest value concentration appears.
- Use the CLV tiers to support marketing budget allocation, loyalty design, and discount strategy.
- Keep the probabilistic CLV model for strategic planning and use XGBoost where a more accurate short-term spend forecast is needed.

Limitations and Next Steps

The model is based on historical transaction behaviour, so it may not fully capture changes in seasonality, promotions, or product availability. A future version could include churn labels, customer demographics, campaign data, and product category features to improve the quality of the recommendations.

A useful next step would be to turn this analysis into an operational dashboard so business teams can track high-value cohorts, tier movement, and forecast changes over time. Overall, the project shows how transaction data can be turned into an actionable CLV framework for retention and growth decisions.